Kamaksha Shekhawat

Machine Learning Engineer | Data Engineer | MSc Artificial Intelligence

London, UK | kamakshaofc@gmail.com | Kamaksha Shekhawat | Kamaksha Shekhawat | Portfolio

PROFESSIONAL SUMMARY

Machine learning and data engineer with nearly 4 years of production experience at Tata Consultancy Services and an MSc in Artificial Intelligence from Aston University (2026). Builds and optimises ETL/ELT data pipelines and large-scale SQL on Snowflake and Google BigQuery, and develops end-to-end machine learning pipelines in Python and scikit-learn, from feature engineering and model training to evaluation and hyperparameter tuning. Hands-on with Natural Language Processing (NLP), Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG). Seeking graduate Machine Learning Engineer, Data Scientist and Data Engineer roles in the UK.

TECHNICAL SKILLS

Programming: Python, SQL, Java, C/C++, JavaScript, HTML, CSS
Machine Learning & AI: Supervised and Unsupervised Learning, Classification, Regression, Clustering, Neural Networks, Deep Learning, Natural Language Processing (NLP), Text Classification, Named Entity Recognition (NER), TF-IDF, Doc2Vec, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Feature Engineering, Model Evaluation, Hyperparameter Tuning
Data Engineering: Extract, Transform, Load (ETL/ELT) Pipelines, Data Modelling, Data Warehousing, Data Quality, Data Validation, Data Reconciliation, Data Migration, SQL Query Optimisation, Partitioning and Clustering, Automated Reporting
Databases & Cloud: Snowflake, Google BigQuery, MySQL, iRAS
Libraries & Frameworks: scikit-learn, Pandas, NumPy, SciPy, Matplotlib, TensorFlow, Keras, NLTK, spaCy, Gensim, SentenceTransformers, Gradio, BeautifulSoup
Tools: Git, GitHub, Jupyter Notebook, Linux, VS Code, MATLAB, CI/CD (basic)
Methodologies: Agile (Scrum), SDLC, ASPICE V-Model, CRISP-DM, Unit Testing, Integration Testing, Requirements Analysis

PROFESSIONAL EXPERIENCE

Data Engineer | *Tata Consultancy Services (Client: DNA Finland, national telecom provider) | India* **Sept 2022 – July 2025**

- Built and enhanced ETL pipelines for monthly STC and non-STC subscription reporting, **cutting processing time by ~40%** and eliminating manual reconciliation across a 3.7M+ subscriber dataset.
- Optimised large-scale SQL transformations in Snowflake and Google BigQuery over millions of telecom subscription and event records, **reducing average query time from ~18 to ~6 minutes (67%)** through clustering, partitioning and query tuning.
- Supported the migration of reporting and analytics workloads from Snowflake to BigQuery (reporting tables, SQL transformations, automated workflows and production datasets), performing large-scale data validation and reconciliation to protect reporting accuracy and business continuity.
- Ran Revenue Assurance (RA) workflows in iRAS, identifying discrepancies that helped **prevent an estimated €200K in monthly revenue leakage**; built analytics to detect ineligible employee-benefit and family-coupon usage across thousands of subscription records.
- Automated Incremental Event Value (IEV) reporting, **reducing turnaround from 2 days to same-day delivery**; prepared and validated datasets for subscription lifecycle analysis, anomaly detection and business intelligence.
- Collaborated with business stakeholders, reporting teams and data engineers to troubleshoot production issues and deliver enterprise reporting, communicating technical findings to non-technical stakeholders.

Automotive Software Engineer | *Tata Consultancy Services (Client: Nissan Japan) | India* **Sept 2021 – Sept 2022**

- Developed and deployed safety-critical ProPILOT features in C++ for embedded automotive systems across the full ASPICE V-model lifecycle, from design through validation.
- Reduced defect escape rate by implementing unit and GUI test automation, contributing to ASPICE Level 2 compliance.
- Optimised real-time sensor fusion algorithms, **improving system response latency by ~15%** with no hardware change.
- Authored Software Requirements Specification (SRS) and Software Detailed Design (SDD) documents and managed client change requests through to implementation.
- Mentored and trained 8 graduate hires in MATLAB, supporting onboarding and team ramp-up.

PROJECTS

Trading Card Price Prediction | *Python, scikit-learn, Pandas, Matplotlib, Jupyter*

- Built an end-to-end machine learning pipeline on ~34K Pokémon cards (cleaned from 40K+ raw records), covering deduplication, a log-transformed target and feature engineering (release era, rarity/holo flags, variants, popularity tier, scarcity score).
- Integrated extrinsic market signals from scraped PSA graded-population data via a variant-aware join (71.6% match rate).
- Benchmarked Random Forest, Gradient Boosting, Linear Regression and MLP against a naive baseline with leakage-safe preprocessing; **Random Forest R^2 improved from 0.854 to 0.876** (log scale) after adding PSA features.
- Validated on an unseen-set holdout (**R^2 0.923**) and a grouped split, with price-band error and feature-importance analysis.

ArXiv Research Platform: NLP and ML Pipeline | *Python, NLP, TF-IDF, Doc2Vec, LLMs, RAG, Gradio, SentenceTransformers*

- Processed and analysed 50,000+ ArXiv papers across 6 research domains using NLP preprocessing, TF-IDF and Doc2Vec feature representations.
- Trained 3 machine learning classifiers across 4 feature configurations, achieving up to **94.5% classification accuracy**.
- Implemented extractive summarisation, LLM-based classification, title generation and Retrieval-Augmented Generation (RAG) workflows.
- Built a Gradio NLP assistant with form-style and chatbot interfaces for real-time classification and summarisation.

Bank Marketing Prediction: End-to-End ML Pipeline | *Python, scikit-learn, Pandas, Jupyter*

- Designed an end-to-end machine learning and data pipeline on the UCI Bank Marketing dataset (~45K records), including data validation, feature engineering, encoding, scaling and model evaluation.
- Benchmarked 6 models across classification and clustering, achieving an **F1-score of 0.91** with tuned classifiers
- Segmented customers into 4 behavioural cohorts using K-Means and hierarchical clustering.

EDUCATION

MSc Artificial Intelligence – Aston University, Birmingham, UK **Sep 2025 – Oct 2026**
Modules: Machine Learning, Deep Learning, Natural Language Processing, Bayesian Analysis, Computer Vision, Robotics and Autonomous Systems

Bachelor of Computer Applications (BCA), 76% – Lachoo Memorial College, Jodhpur, India **Jul 2018 – Sep 2021**
Modules: Data Structures, Database Management Systems (DBMS), Object-Oriented Programming (OOP), Software Engineering, Computer Networks